

HackUPC 2026: EdgeAI for a Resilient and Greener Barcelona - Information Guide

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1. Introduction

Welcome to **HackUPC 2026**, organized in collaboration with **Arduino**, **Qualcomm**, and the **Universitat Politècnica de Catalunya (UPC)**.

Barcelona is facing increasing climate challenges: pollution, extreme heatwaves, loss of biodiversity, and water scarcity. This hackathon challenges you to use **EdgeAI** to build a more resilient city. By leveraging **Qualcomm** technologies, you will create on-device decision-making systems that monitor, protect, and enhance our urban environment in real-time, reducing dependence on centralized cloud infrastructure.

2. Schedule & Deadlines

- **Hacking Start:** April 24, 2026 – 8:30 PM (after the Opening Ceremony)
- **Tech Workshops:** April 25, 2026 – 10:00 AM to 1:00 PM
- **Hacking Stop:** April 26, 2026 – 9:00 AM
- **MANDATORY Submission Deadline:** April 26, 2026 – 9:15 AM
- **Judging Session:** April 26, 2026 – 10:15 AM to 1:15 PM
- **Closing Ceremony & Awards:** April 26, 2026 – 3:00 PM

3. The Technical Challenge

Build an EdgeAI system using the **Arduino UNO Q** and **Edge Impulse** that can:

- **Collect and process** local sensor or camera data on-device.
- **Run inference locally** (classification, object detection, anomaly detection, audio detection, or small LLMs and VLMs).
- **Make autonomous decisions** and trigger local actions.
- **Operate reliably** under real-world constraints (limited power, memory, and connectivity).

The goal is to create practical, impactful solutions that address Barcelona's urban resilience and green challenges.

4. Hardware Kit & Resources

Each team will receive a specialized **Edge AI Kit**:

- **Brain:** Arduino UNO Q.
- **Vision:** USB Web Camera.
- **Sensors and Actuators:** Arduino Modulino Nodes (*Movement, Distance, Thermo, Vibro, Light, Knob, Button, Buzzer and Pixels*) & QWIIC cables.
- **Connectivity:** USB Hub, Power Supply, and USB-C cables.
- **Software:** Arduino App Lab IDE & Edge Impulse Studio.

5. Project Submission and Guidelines

To be considered for evaluation, every team **MUST** complete a **three-step submission process**. Failure to complete any of these steps will result in disqualification from the judging process.

5.1 Devpost

Register your project on the official HackUPC Devpost page:

<https://hackupc-2026.devpost.com/>

- Use Devpost to submit your project summary, team members, and the mandatory links to your GitHub repository and Arduino Project Hub page.

5.2 GitHub Repository

Upload your complete project to a public GitHub account.

- **App Lab Application:** All files including `main.py`, `sketch.ino`, and YAML configs.
- **Documentation:** A clear and comprehensive `README.md`.
- **License:** A `LICENSE.txt` file (Mozilla Public License 2.0).

Reference Example: [GitHub: Greetings from Uno Q](#)

5.3 Arduino Project Hub

Create a detailed project page on the **Arduino Project Hub** (<https://projecthub.arduino.cc/>).

- **Mandatory Naming Convention:** [HackUPC2026] - Your Project Name
- **Required Content Structure:**
 1. **Project Description:** A high-level overview of the urban problem solved.
 2. **Hardware Lineup:** Detailed list of hardware (UNO Q, Modulinos, Camera).
 3. **The AI Model:** Information on the model used (e.g., FOMO via Edge Impulse).
 4. **Software Architecture:** How the application logic processes data locally.
 5. **Visuals & Media:** High-quality images and a short demonstration video.
 6. **Github Link:** A direct link to your repository.

Reference Example: [Project Hub: Greetings from Uno Q](#)

6. Application Areas & Project Ideas

To spark your creativity, here are the core challenges for Barcelona's urban resilience. These examples show how Edge AI can solve real problems.

Remember, these are just starting points... **space to your imagination!**

Surviving Heatwaves

Protect citizens from extreme heat in public spaces.

- **Heatwave Alerting System:** Detect dangerous heat indexes using Modolino sensors and provide real-time recommendations via local LLMs on the UNO Q.
- **Intelligent Cooling:** Automatically activate fans or misting systems in public waiting areas only when extreme temperatures are detected.
- ...

Water Conservation

Reduce water waste in parks, streets, and at home.

- **Leak & Waste Detection:** Use audio anomaly detection to identify pipe bursts in public fountains or computer vision to shut off beach showers left running unnecessarily.
- **Smart Irrigation:** Use FOMO (Object Detection) to assess plant health and soil stress, ensuring water is used only when and where the greenery actually needs it.
- ...

Urban and Coastal Cleanliness

Keep streets, parks, and beaches clean using EdgeAI.

- **Smart Bin & Dumping Monitor:** Use cameras to detect when bins are full or to identify illegal waste dumping in real-time.
- **Beach & Street Optimizer:** Use vision to identify plastic waste on the sand or measure dirt levels on streets via bike-mounted cameras to optimize cleaning routes.
- ...

7. Judging Criteria

Criteria	Focus
Creativity & Relevance	Originality of the solution for Barcelona's green challenges.
Technical Execution	Efficiency of the EdgeAI model (latency, accuracy).
Impact & Scalability	Potential for real-world deployment in public spaces.

8. Resources & Support

- **Arduino UNO Q Docs:**
<https://docs.arduino.cc/hardware/uno-q/>
- **Arduino App Lab Guide:**
<https://docs.arduino.cc/software/app-lab/>
- **Arduino Modulino Getting Started:**
<https://docs.arduino.cc/tutorials/modulino-thermo/how-general/>
- **Edge Impulse & Arduino App Lab:**
<https://docs.edgeimpulse.com/hardware/deployments/run-arduino-app-lab>
- **Official Arduino Forum:**
<https://forum.arduino.cc/>

Note: Technical mentors from Arduino, Qualcomm and Edge Impulse will be available on-site during the entire event to assist with hardware integration and model optimization.

Good luck! Let's build a smarter and greener Barcelona together.